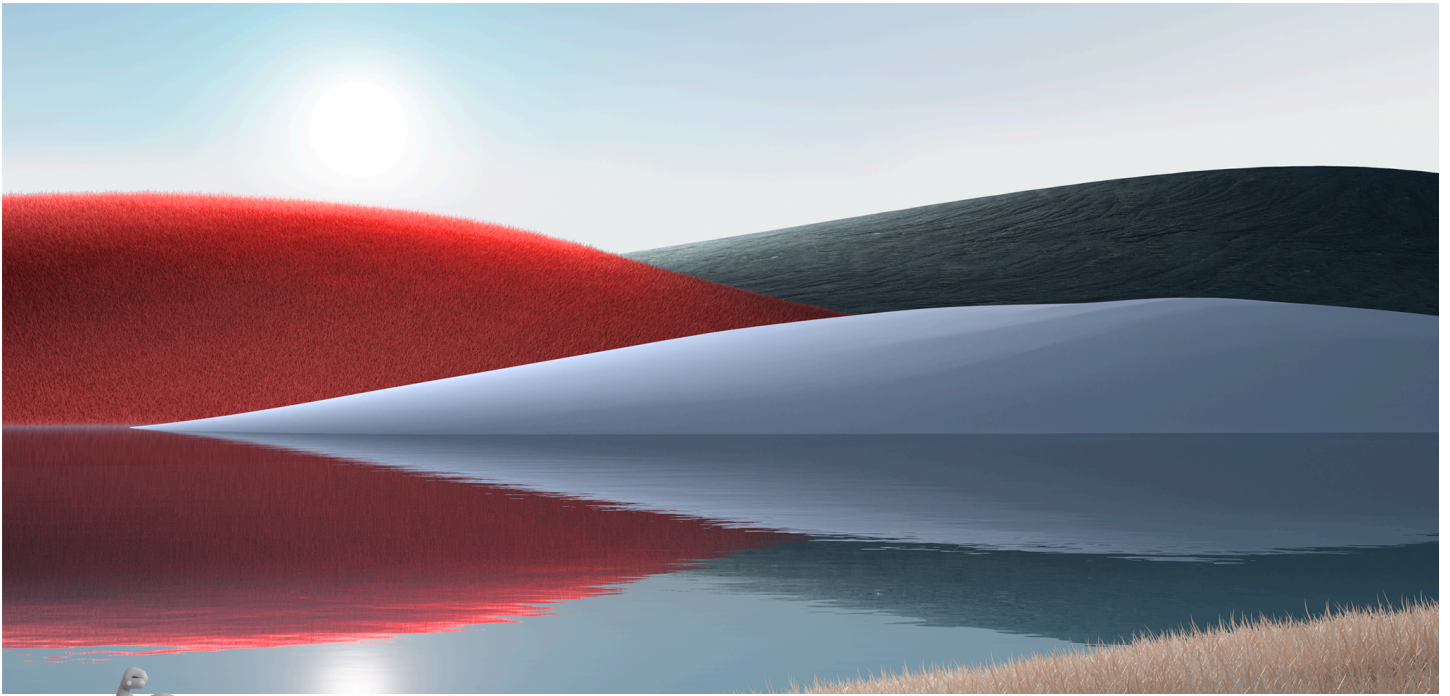




Prostate Cancer 3D

MILESTONES

	Work Area	Owner	Progress	Blockers	Checklist	Completion Date	Notes
1	Ensure Training of the model can work and check the key computing resources needed	PA	Completed	SUFYAN: Should the internet fail, please transfer me the app with the updated requirements.txt I will send it to chroma to run			Train using AWS OR HIVE OR CHROMA
2	Set up proper training script to be ported over to AWS	Anson	Completed		✓ build inference pipeline to generate 3d segmentation — nifty		Also need to quickly work on the inference pipeline to generate niftti files for the image segmentation and also ensure is validated on the original image size
3	Prepare Commercial cloud account for Asher	Sufyan	Not started		- To create another IAM account for Asher - Set proper rights - Identify all the services needed before executing - Brief through the plan and confirm		Populate the list of things in this board as separate items to manage and clear
4	Need to setup billing tracker service first	Asher	Completed		Enabled AWS cost optimizer already		Set budget and monitor trajectory Populate the list of things in this board as separate items to manage and clear

5	Set up AWS services for training	Asher	In progress		1. set up IAM users 2. choose appropriate Sagemaker environments to run training script 3. familiarise with RRU-net architecture		Need help to report Sufyan every time a service is being used and why. Populate the list of things in this board as separate items to manage and clear
6	Port over codes from local to AWS	Asher	Not started				
7	Run Training in AWS	Asher/ Anson	In progress				
8	Decision making process in porting code over	Asher/ Anson	In progress		1. Can the dataset be stored in an S3? 2. Distributed computing to increase efficiency? 3. which GPU instance should be used?		
9	learn deep learning	Asher	In progress		✓ Learn forward pass/backpropagation ✓ CNN/UNet architecture ✓ RRUNet3D model in image segmentation		
10	practice pytorch deep learning on sagemaker studio	Asher	In progress		github repo sagemaker 101 workshop walkthrough: ✓ custom script demos ✓ migration challenge		

11	sagemaker cost	AG	Not started		check how to reduce costs during training (GPU instances for distributed computing, warm pooling, persistent cache)	☐ Update cost tracking using this link in the notes	AWS Billing Tracker.xlsx
12	sagemaker efficiency: distributed computing	Asher	Not started		tflops & speed/power vs cost (update here which instances i'll use)		
13	training vs inference	Asher	Not started		design principles to consider: do i put training script together with/separate from inference? do i need to consider batch inference? (sagemaker batch transform)		
14							

Tasks

	Work area	Notes	Owner	Status	End date	Blocker
1	Monai SDK action items: 1. setup linux environment - Linux mounted on top of windows using WSL 2 2. Install Nvidia Driver and CUDA Toolkit 3. Install WSL Extension for IDE 4. Mount Local files onto the remote workspace	27 Oct Unable to load image Try reloading the page	Phua Anson (SingHealth - SGH)	Completed		
2	Explore Weasis app to load Dicom and Dicom RTSTRUCT Files	29 October 4pm meeting	Phua Anson (SingHealth - SGH)	Completed		
3	Talk to Anh and work with him to get the dicom series description out.	29 October 4pm meeting	Phua Anson (SingHealth - SGH)	Completed		✔ Wait till Anh is back from holiday
4	look at inference and prostate model from monai-sdk (After WSL is setup) the nvidia/nih one and do the same	Dataset provided by the NIH Cancer Dataset, ProstateX challenge, tested on ProstateX-004 patient MRI Data gonna refactor their models nicely, just add_flow.	Phua Anson (SingHealth - SGH)	Completed		
5	RE-Modify Inference Pipeline to output RTSTRUCT file	Dr Anh investigated and highlighted my RTSTRUCT was formatted wrongly, suggested using another dataset, and using his prompt	Phua Anson (SingHealth - SGH)	Completed		
6	To learn how to refactor code also	refactoring. Guru martin fowler official refactoring site freecodecamp - code refactoring best practices	Phua Anson (SingHealth - SGH)	Completed		
7	Give a summary of what has been done so far		Phua Anson (SingHealth - SGH)	Completed		

8	Redo Combined inference pipeline to ensure corrected RTSTRUCT Exports with NEW Dicom Data (from nvidia)		Phua Anson (SingHealth - SGH)	Completed		
9	Further clarifications sought with Anh and team on DWI vs highb volumes		Phua Anson (SingHealth - SGH)	Completed		
10	how nvidia is being trained on and what?		Phua Anson (SingHealth - SGH)	Completed		
11	To refer to the inference script, need dissect input and output for each model until reach the end.		Phua Anson (SingHealth - SGH)	Completed		
12	To test the Prostate-X dataset test data (choose one patient) on open source prostate158 app		Phua Anson (SingHealth - SGH)	Completed		
13	Verify segmentation mask and validity of Prostate-X prostate and lesion masks (ground truth) from https://github.com/rcuocolo/PROSTATEx_masks	Verified already.	Phua Anson (SingHealth - SGH)	Completed		
14	Setup prostate-monai-app repo remote via suregohome		Phua Anson (SingHealth - SGH)	Completed		

15	[BACKLOG] RTSTRUCT Conversion from Semantic Segmentation Masks (in niftii format)	I've managed to successfully convert the PROSTATEx NiftI lesion segmentation into an RTSTRUCT that aligns correctly with the T2 DICOM series for ProstateX-0004. In 3D Slicer, the RTSTRUCT contour (red) now overlaps perfectly with the original NiftI lesion segmentation (green) on the T2 series, confirming that the geometry and orientation are consistent. (1) RTSTRUCT Conversion from Nifti Segmentation Mask				
16	Create relevant folders for training pipeline in prostate monai app					
17	Iron out train-test-validation split of the 204 prostate-x data					
18	Seek input for Dr Anh on whether to flood the holes for RTSTRUCT output or not	Yes conclusion from Anh is to flood the holes as he explained " ah yes, RT struct assume there is no hole in a contour" yes as in to flood the holes				
19	Setup normal organ training pipeline and scripts - Version 1 is based on previous prostate158 with help of cursor AI					
20	Prepare train.csv, test.csv, valid.csv for training					
21	5-fold training pipeline for Model training (Multi-class zonal prostate segmentation)	fold 0 dice 0.7288 check if got any post-processing cleaning (such as removing the noise)				
22	Batch Processing with RTSTRUCT output for App inference	rtutils and nibabel and scipy (filling binary holes)				

23	find out .pth, .ts, .pt		Phua Anson (SingHealth - SGH)	Completed		
24	Verify prostate output		Phua Anson (SingHealth - SGH)	Completed		
25	Critical bug fix — multi-class organ segmentation Prostate Inference Pipeline double check again,	After reviewing for more than a day wrecking my brains and getting bamboozled by AI, key takeaway is: For multi-class, setting the asdiscreted threshold to None and setting argmax is True will suffice as argmax already decides for each voxel which is the winning class (background/TZ/PZ)	Phua Anson (SingHealth - SGH)	Completed		
26	Validation and testing	(1) Sprint achievements for the Week of 8 Dec 2025	Phua Anson (SingHealth - SGH)	Completed		
27	Documentation of what has been done so far	(1) Sprint achievements for the Week of 8 Dec 2025	Phua Anson (SingHealth - SGH)	Completed		
28	Sanity check: Why model.ts save requires a trace function beforehand and an example input?	Already noted from documentation this is the procedure to capture the computation graph.	Phua Anson (SingHealth - SGH)	Completed		
29	Sanity check: Model.ts = model.pth???	To evaluate DICE on BINARY whole-gland mask (TZ + PZ = prostate gland) Hence, TZ+PZ collapsed to foreground. Labels remain multi-class on disk but the evaluation binarises them with labels > 0. Dice metric is set to include_background=False so it reports the foreground (whole gland) dice DICE:0.8515	Phua Anson (SingHealth - SGH)	Completed		

30	Tumor training (Sanity check)	so far dice is around 0.21, each fold only 20 epoch. haven't really train yet and was figuring our the training pipeline, to update once all 5 folds done. So far 60 epochs per fold done on 5 folds : Fold 0: 0.2024 Fold 1: 0.1816 Fold 2: 0.2414 Fold 3: 0.2456 Fold 4: 0.2377	PA Phua Anson (SingHealth - SGH)	Completed		
31	Tumour Training (To train using ADC Tumour ground truth instead) Purpose: I want to see whether using ADC Tumour ground truth will give a better dice score first for 60 epochs before committing to training. Also important	Not much diff. Fold 0: 0.2082 Fold 1: 0.181633 Fold 2: 0.2414 Fold 3: 0.2393 Fold 4: 0.1837	PA Phua Anson (SingHealth - SGH)	Completed		
32	Tumour Training (Lock-in)		PA Phua Anson (SingHealth - SGH)	In progress		
33	lesion : dependent or not dependent of prostate segmentation					
34	Verify tumour output		PA Phua Anson (SingHealth - SGH)	Completed		

35	Questions to ask felicia/Deepak		Phua Anson (SingHealth - SGH) PA Mohammad Sufyan Bin Amirhazam (SingHealth - SGH)	Completed	Mo...	
36	Test Re-training for Tumour (Combine the ground truth mask of T2 and ADC, as updated by Dr Anh)	VAL DICE : 0.32 highest	Phua Anson (SingHealth - SGH) PA	Completed	Mo...	✓ Rushing on D3, only left with 2-3 weeks :))
37	To test out whether can integrate SGH Sample Data (4 sets given by Jun Han) on our Monai App		Phua Anson (SingHealth - SGH) PA	Completed	Sun,...	
38	Template		Phua Anson (SingHealth - SGH) PA		Tue...	
39	Excel sheet of the 4 sets of data (what is needdd)		Phua Anson (SingHealth - SGH) PA		Mo...	
40	Refactor the open source repo	i probably will refactor the prostate-monai-app instead since im gonna be building the training pipeline from scratch	Phua Anson (SingHealth - SGH) PA	Blocked		

Project Timeline

Add based on data science workflow!

Visualize my project in phases (how to work with)

	Task	Assigned to	Due date	Bucket
1	☐ [Project Understanding] - Confirm problem statement of project/success metrics and roles - E-learning of Deep Learning	Phua Anson (SingHealth - SGH)	Sun, Jun 1, 2025	Completed
2	☐ [Data Understanding] Data Readiness: Confirmation of Open-source Data to use, EDA of imaging data	Phua Anson (SingHealth - SGH)	Sun, Jun 1, 2025	Completed
3	☐ Modelling: Baseline modelling of Prostate and Tumour Project		Sun, Aug 31, 2025	Completed
4	☐ Model Evaluation and Iteration: Create prostate and anatomy training notebook, with 10 cases overlay review done via 3DSlicer		Fri, Oct 31, 2025	Completed
5	☐ [Packaging] Redo Combined inference pipeline to ensure corrected RTSTRUCT Exports with NEW Dicom Data (from nvidia)		Mon, Nov 10, 2025	In Progress
6	☐ [Refactoring] Refactor the codes in this project		Mon, Nov 17, 2025	To do
7	☐ Integration into Monai SDK		Mon, Dec 1, 2025	To do
8	☐ [Model RE-Evaluation and RE-Iteration] Train with SGH Data		Wed, Dec 31, 2025	To do
9	☐ [Deployment] Work with biobot and Anh on this			To do
10	☐ [Monitoring] Check for Data Completeness; geometry consistency, Image Drift??			To do

Questions for deepak/felicia

CONTEXT:

- Dr.Anh has managed to extract a good number of studies, he will get in touch with DyGCMIO to anonymise them, in the meantime, he is requesting DDS to review the CARPL's implementation of the NIH Model, which is this app and model on our server as he wants to send the deidentified MRI prostate to that model to generate contour + tumour segmentation from the CARPL team's model which we will use to train our model using this app.
- They have deployed in RAPIER server which deepak (from carpl) has showed me how to access. He needs my help to investigate the NIH Prostate model so we can ingest dr Anh's data.
- Dr Anh feels it's best we build the docker by myself, so he predicts that we must optimise the sequence selection to adapt to our own studies., to access the RAPIER server, he will get felicia to facilitate.

WE LOOK LOOK THROUGH ENTIRE SETUP AND CONFIGURATION AND INVESTIGATE THE GENERAL VOLUME SIZE WHAT DIMENSIONS OF T2, ADC, HIGHB, HOW VOLUMES SELECTED, FOR TRAINING HOW SHOULD FILES BE ORGANISED, WHAT SERIES SELECTION, AND ASK AS MANY KEY QUESTIONS TO DEEPAK FROM CARPL AS MUCH AS POSSIBLE.

What I recall from Deepak:

- He explained how CARPL receives the MRI volumes from PACs (iirc), and CARPL sends it to the Rapier box for the modelling and inference then send to the radiologist for review
- RAPIER is based on Linux system, and based on the NIH App
- Deepak explained and showed that Batch Inference is possible
- Deepak mentioned that for the inference to work, put the multiple MRI volumes into an "input" folder, and then after inferencing the images and all will be sent to the "output" folder. He showed me using the Weasis app.
- Weasis app used in his own laptop
 - Input - folder
 - inference output - in nifti
 - output - in harddisk - in nifti
 - The test case images used - what are they

Also some login details:

Here are the simple steps you can follow to help us get the data from the Rapier system. We'll be using this data to create a reference document for you.

Please let me know if you face any issues - we can quickly get on a Teams call to sort it out.

Steps:

1. Log in using the following details:

Username: anson (this should already appear by default)

Password: anson@monai

2. Insert a USB drive or any external storage device into the server.

3. Open the file manager and go to this path: /home/anson

4. Copy all the files in the /home/anson folder.

5. Zip the copied files and send them to us.

Thank you for your help!

We need to upload manually from sharefolder - hard disk - then into CARPL

- I recall only seeing Deepak show me BINARY prostate output on weasis app.
- Deepak mentioned it is based on the Series selector ??
- Deepak also said CARPL side can also help to deidentify and anonymise our SGH Data?
- To use Encrypted hard disk to take out the image segmentations and images and put it into our local training laptop for modelling

Series Selection (T2 / ADC / HIGHB)

```
Rules_T2 = """"{
  "selections": [{
    "name": "t2",
    "conditions": {
      "Modality": "MR",
      "ImageType": ["ORIGINAL", "PRIMARY"],
      "SeriesDescription": "((?=AX T2)|...[(?=T2W_TSE_ax zoom PSS Refoc))"
    }
  }]
}""""

...

Rules_ADC = """"{
```



```

"selections": [{
  "name": "adc",
  "conditions": {
    "Modality": "MR",
    "ImageType": ["DIFFUSION", "ADC"],
    "SeriesDescription": "((?=ADC (10^-6 mm^2/s))|...|(?=ep2d-advdiff-3Scan-4bval_spair_511b_ADC))"
  }
}]
...
Rules_HIGHB = """{
  "selections": [{
    "name": "highb",
    "conditions": {
      "Modality": "MR",
      "ImageType": ["DIFFUSION", "TRACEW"],
      "SeriesDescription": "((?=AX DWI)|...|(?=sDWI_b2000))"
    }
  }]
}"""

```

This looks hard-coded. We need to expand/update the regex for our local naming conventions, otherwise the series won't be selected and the pipeline will fail.

Hi Anh, we both discussed, we have some questions below:

1. We need a sample to test the current tag rules using regex used in RAPIER, we haven't ascertain how good it is in taking out the right (T2/ADC/HighB). We suggest that we can get a few batches first to test while waiting for the rest to be de-ied.
2. For HIGHB is there a range to look out for? I recall like B value being 800 or something to be DWI?
3. For the inference by NIH model in CARPL, currently is just binary - and its sufficient to get the tumor after. The Multi-class prostate segmentation can only be done using open-source data, if ever need for multi class fine-tuning using SGH data, will need your help to re-label prostate class into transitional/peripheral zone.
4. Was there a confirmed plan on how CARPL NIH model takes in data from? Is it from RISPACS straight (this was what Deepak mentioned that time when introducing how input data comes in) , or the de-id data from the shared folder pushed to CARPL via network or HDD? If not sure, its k, we can ask them later.

QUESTIONS

- How do they resolve multiple candidate series per modality (T2/ADC/HIGHB)? Is there tie-breaking logic? - can ask
- Any expected metadata fields that must be preserved in anonymisation?
- Can we access the code base in the desktop with felicia on site?
- Were there any changes to preprocessing steps? which part was changed? Help run through in this meeting if possible?
- Dont mind showing us again on how batch processing happens, and where the input and output is.
- How many batches of images during inference?
- Can Felicia be able to run the codes without us inside, just help to transfer over from sharefolder into CARPL for inference.
 - Is there a limit in how much data (input and output) can be stored in the desktop? or is it seating somewhere else?
 - For the SGH data that has been de-id, it would be in the sharedfolder. not from rispacs. So for this setup that CARPL has made, would the pipeline require de-id image? or straight from RISPACS?
 - if de-id, would that be ok if images were transferred from intranet sharedfolder, then pass through intranet into CARPL or better to load manually into desktop?

Pre-processing and geometry method:

- Do you resample to 1.0 mm or 0.5 mm in their deployment? Any differences from this repo?
 - current was 1.0mm
- Any preprocessing beyond what's in the code (N4 bias correction, denoise, intensity normalization, etc.)?
 - got changes done on the preprocessing
- Do they force a specific orientation (RAS)? If not, how do they handle orientation mismatches?

MODELLING:

- Threshold used for lesion mask (0.63 here) ? same on RAPIER?
- Expected input folder structure on RAPIER (study-level vs series-level)? double check what deepak mentioned that just requires input and output folders
 - Models same as the one in NVIDIA/NIH?
 - Multi-class prostate segmentation or just binary?? - Anh
 - Accounting for failures in selection/mis-routed series cases, how can we detect, automated checks on their end have or we need to DIY?

Raw DICOM dimensions for ProstateX-0004

From data/raw/ProstateX-0004:

Likely T2 series (matches "t2_tse_tra" rule in app):

- t2_tse_tra
- Rows×Cols: 384×384
- PixelSpacing: [0.5, 0.5] mm
- SliceThickness: 4.5 mm
- SpacingBetweenSlices: 4.5 mm
- NumInstances: 19

Diffusion / High-b candidates:

- ep2d_diff_tra_DYNDIST_MIX (TRACEW, DIFFUSION)
- Rows×Cols: 128×84
- PixelSpacing: [2, 2] mm
- SliceThickness: 4.5 mm
- SpacingBetweenSlices: 4.5 mm
- NumInstances: 57
- ep2d_diff_tra_DYNDIST_MIXCALC_BVAL (TRACEW, DIFFUSION)
- Rows×Cols: 128×84
- PixelSpacing: [2, 2] mm
- SliceThickness: 4.5 mm
- SpacingBetweenSlices: 4.5 mm
- NumInstances: 19

ADC candidate:

- ep2d_diff_tra_DYNDIST_MIX_ADC (ADC, DIFFUSION)
- Rows×Cols: 128×84
- PixelSpacing: [2, 2] mm
- SliceThickness: 4.5 mm

- SpacingBetweenSlices: 4.5 mm
- NumInstances: 19

Other series present (e.g., t2_tse_sag, t2_tse_cor, multiple tfl_dyn_fast_tra...) are **not** selected by the current regex rules.

How ingestion/selection works

- The pipeline uses `DICOMDataLoaderOperator` → `DICOMSeriesSelectorOperator` → `DICOMSeriesToVolumeOperator` for each modality (T2, ADC, HIGHB).
- Selection is **regex-based** on **DICOM tags**: Modality, ImageType, and SeriesDescription.
- The rules are **hard-coded** in `prostate_mri_lesion_seg_app/app.py` under `Rules_T2`, `Rules_ADC`, and `Rules_HIGHB`.

How volumes are standardized after selection

- T2 is reoriented to **RAS** and resampled to **1.0 mm isotropic** for organ segmentation.
- ADC and HIGHB are first resampled to **match T2 geometry**, then all three channels are resampled to **0.5 mm isotropic** for lesion segmentation/classification.
- ROI is cropped around the prostate mask with **32-voxel margin**.

Volume Size / Dimensions (T2, ADC, HIGHB)

There is **no fixed voxel dimension** required. The pipeline rescales/resamples inputs in specific ways:

Organ Segmentation (T2 only)

- Reorients to RAS
- Resamples to **1.0 mm isotropic**
- Inference uses sliding window ROI size (**128, 128, 16**)

```
Orientationd(keys=my_key, axcodes="RAS"), Spacingd(keys=my_key, pixdim=[1.0, 1.0, 1.0], mode=
["bilinear"]), ... infer_operator = MonaiSegInferenceOperator(..., roi_size=(128, 128, 16),
overlap=0.5, sw_batch_size=4)
```

Lesion Segmentation (T2/ADC/HIGHB + organ mask)

- ADC/HIGHB are resampled to **match T2 geometry** (size/spacing/origin/direction)
- All channels then resampled to **0.5 mm isotropic**
- ROI cropped based on prostate mask + **32-voxel margin**
- Inference done in patches that are multiples of 32

```
img_resampled = sitk.Resample(... t2_sitk.GetSize(), ... t2_sitk.GetSpacing(), ...) ... spacing_target
= (0.5, 0.5, 0.5) shape_target = [int(round(nda_shape[i] * spacing_orig[i] / spacing_target[i])) for i
in range(3)] ... margin = 32 bbox = bbox2_3D(nda_wp_resize) ... nda_resize_roi = nda_resize[:,
bbox_new[0]:bbox_new[1], ...]
```

Lesion Classification

- Resamples all channels + masks to **0.5 mm isotropic**
- Crops **64×64×64** around each lesion component
- **Raw DICOM sizes are not fixed**; they vary by site.
- The pipeline **standardizes spacing**, not fixed voxel counts.
- Resampled volume shapes depend on original spacing and FOV.

Training Data Organization

Prostate Training expects NIfTI in

`data/preprocessed/` with specific subfolders.

```
t2_dir: data/preprocessed/t2
```

```
label_dir: data/preprocessed/organ_masks
```

Labels must be **multi-class** (0=background, 1=TZ, 2=PZ).

So i foresee we take out the the segmentation and images from OUTPUT and put inside these pre-processed files.

Lesion training

```
t2_dir: data/preprocessed/t2
```

```
adc_dir: data/preprocessed/adc
```

```
highb_dir: data/preprocessed/highb
```

```
organ_mask_dir: data/preprocessed/organ_masks
```

```
lesion_mask_dir: data/preprocessed/lesion_masks
```

SUMMARY OF PROJECT

▼ AWS account

AWS Commercial Cloud

Account ID: 345594598345

Root: dept.of.data.science@sgh.com.sg

Root password: DD\$sgh.#01

Follow-ups

	Column 1	Column 2
1	AG Asher Lim Guojun (SGH) Kindly provide the link to the one note once you're done. Please seek advise from Anson where to find the notebook. Ideally should put at here - make another subpage AWS SageMaker (Web view)	Asher @ HealthSG AG Asher Lim Guojun (SGH) During presentation, please reference back to this OneNote to go through your journey necessary steps you've taken. Dont have to do live demo in aws service for all (like creating IAM user)
2	PA Phua Anson (SingHealth - SGH) Please let me know your future plans this week for this project.	Plan for this coming week (Under Sprint 5, will check with Asher later see if anything can add or miss out too) - Development of Anatomy Segmentation Model using RRUNET3D (Modify config file) by the end of this week - To verify the anatomy segmentation model with 3DSlicer - To discuss with Asher by this week on how to deploy our tumor/anatomy model into AWS Backlog - Settle the pipeline and Verify the tumor prediction output with 3DSlicer again (After monai.transforms.inverse is applied as advised. Side thought Ok the thing is I realized we have been using other people's code all these while, I was thinking to start doing the model from scratch when I am free. - Sufyan: you can always do so, if you have the right data in and you know the output u need. But with time as a factor, we can discuss this further in call

AG Asher Lim Guojun (SGH) Kindly help provide the link to the write-up here - editable

Documentation: https://entuedu-my.sharepoint.com/:o/r/personal/asher0003_e_ntu_edu_sg/Documents/AWS%20Sagemaker%20Documentation?d=we560bb47602a45eabd37154256875945&csf=1&web=1&e=4ywwFF

Progress on AWS SageMaker Training:

1. Testing codebase in Sagemaker Studio and encountered some bugs:
 - a. Cleaned up requirements.txt file
 - b. Chose Pytorch 2.6 image for Estimator object (previously the estimator was using python version 3.10 which is incompatible with MONAI packages)
 - c. Testing using ml.t3.large instance (CPU only) to minimize cost
 - d. Update documentation

2. Moving forward:

- Want to implement heterogeneous cluster for pre-processing (CPU only) and GPU instance for a training job
- modify code to support distributed training (DDP)
- Incorporate anatomy model into Sagemaker pipeline

Sprint 8 [30/9/25 - 18/10/25]

	☰ Task	📌 Progress	☰ Remarks
1	Explore Weasis app to load Dicom and Dicom RTSTRUCT Files with Anh's Data	In progress	Put on hold to settle with Wan Chee the Healthhub+ deployment meeting
2	Modify Inference Pipeline to output RTSTRUCT file	In progress	Put on hold to settle with Wan Chee the Healthhub+ deployment

Sprint 7 [29/8/25 - 14/9/25]

As updated by Sufyan, Anh's side will be deploying open source model in carpl. Might need to do tweaking for the search for the right images, the 3 types that sufyan told me about.

	☰ Task	📌 Progress	☰ Remarks
1	Get familiar with the contents in the dicom	Completed	
2	Retrain tumor segmentation once again for RRUNET3D	Completed	Looks like for tumor segmentation, RRUNET3D (Dice score 0.36) still pales in comparison to UNET (Dice score: 0.45).
3	Combining anatomy segmenation and tumor segmenation pipeline into one script only.	Completed	Make some tweaks and refactor as well to the code

Few Questions to check with Dr Anh:

I had a demo session for the Monai model, and a couple of clarifying questions to update Simran and Deepak from Carpl

- You mentioned that the studies will be run in a batch manner. Could you share the estimated size of each batch so we can make the necessary adjustments on our side?
- In addition, if possible, could you also let me know the total study volume you're planning to run on the solution?

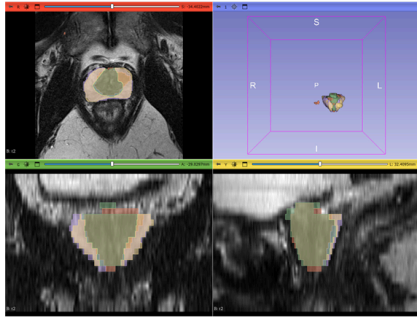
Sprint 6 [11/8/25 - 17/8/25]

	☰ Task	📌 Progress	☰ Remarks
1	To start git-ing and upload to suregohome	Completed	I created a branch and pushed it to remote suregohome p158 repo already. The branch is called <u>feature/inference-pipeline-no-aws</u> Why I created this branch: - refactored my codes and notebook as it was too messy and i wasn't really able to track changes efficiently and effectively - Ensure the pipeline is working properly first w/o rushing to aws Next step is to: Review Asher's AWS pipeline and notebook and integrate to the main branch once ready.
2	Further continue training of ANATOMY (T-ZONE/P-ZONE of prostate) segmentation model using RRUNET	Completed	Latest update as of 12 Aug: At the 436th epoch, the DICE Score stands at a decent 0.7916. At the 500th epoch, the DICE Score stands at 0.7959. To continue iterating till the 1000th epoch.
3	To further verify the inference pipeline for tumor segmentation model with 3DSlicer, with 10 patient cases	Completed	Findings can be found below. Explanation of inference pipeline can be found in Sprint 5.
4	To verify the anatomy segmentation model with 3DSlicer, with the same 10 patient cases as the tumor segmentation model	Completed	Verified and the affine is correct. Overlayed ground truth with model prediction. Tallies well.

Anatomy Segmentation Update

Case Studies - Anatomy Seg DICE SCORE: 0.7959 RRUNET3D MODEL (RED/PURPLE is trained model and Yellow/Green is Ground Truth)

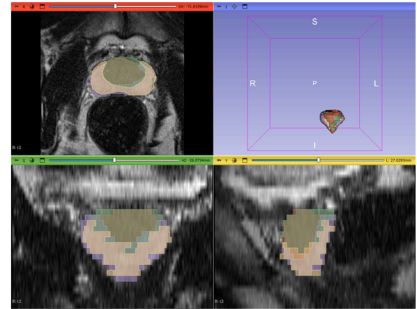
CASE 51 - ANATOMY



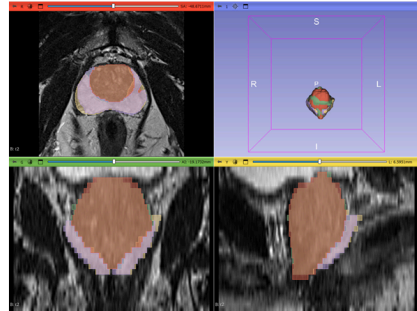
CASE 43 - ANATOMY



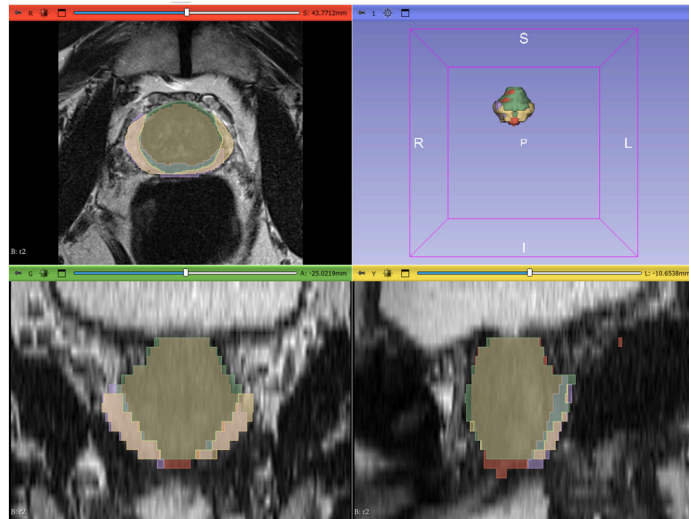
CASE 63 - ANATOMY



CASE 86 - ANATOMY



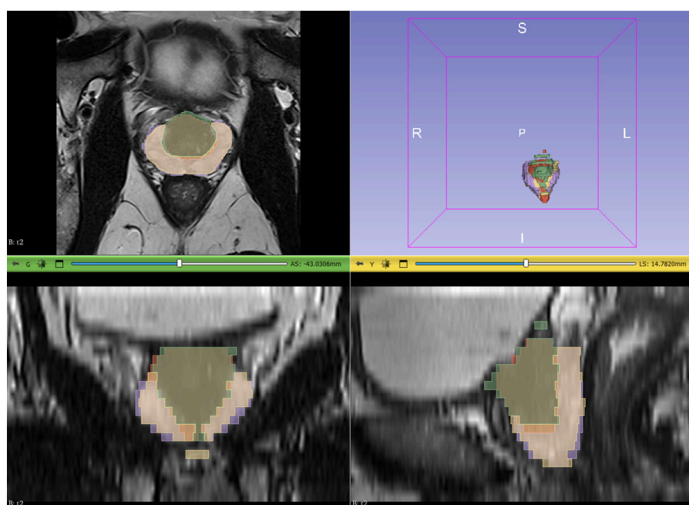
CASE 100 - ANATOMY:



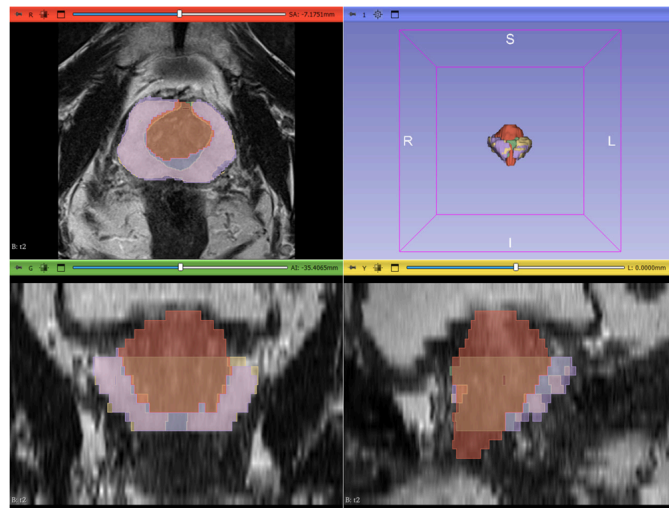
CASE 118 - ANATOMY



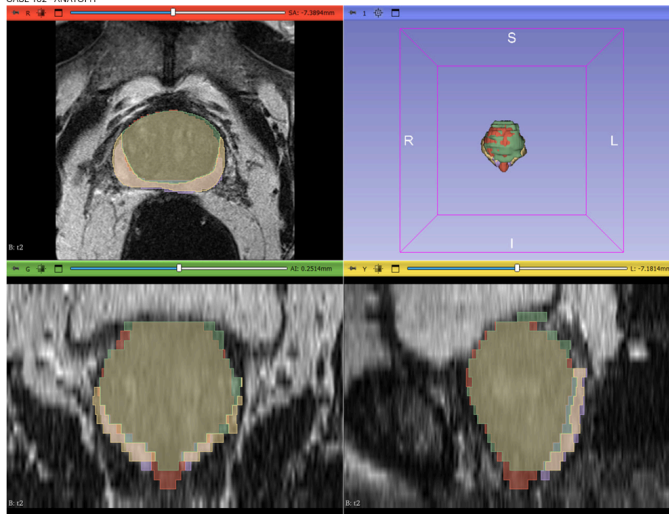
CASE 122 - ANATOMY



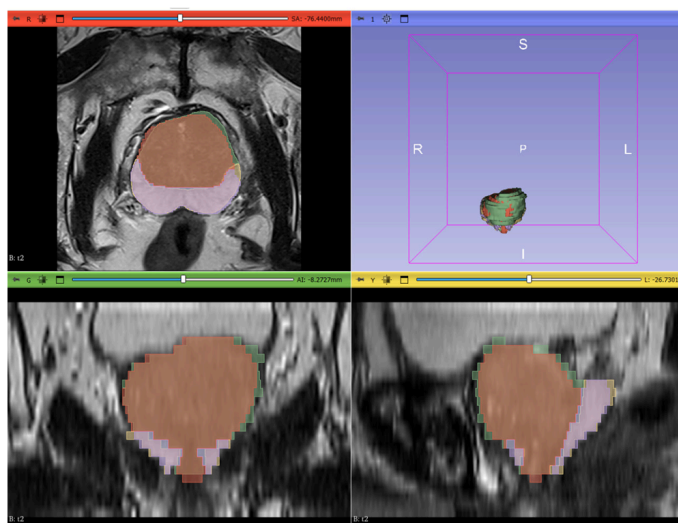
CASE 125 - ANATOMY



CASE 152 - ANATOMY



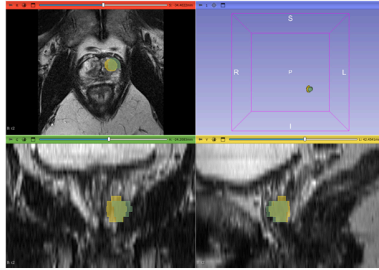
CASE 158 - ANATOMY



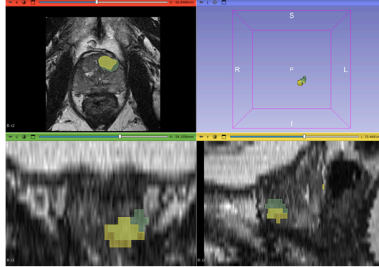
Tumor Segmentation Update

Case Studies - Tumor Seg (Yellow is trained model and Green is Ground Truth)

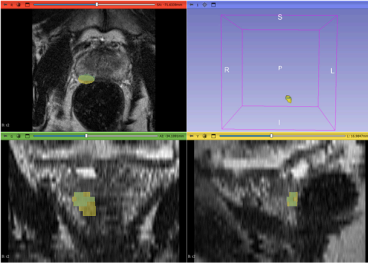
CASE 51:



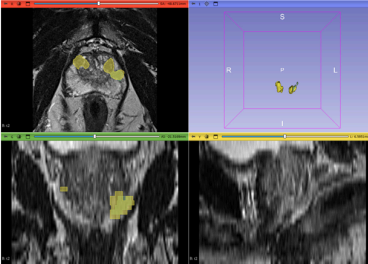
CASE 43:



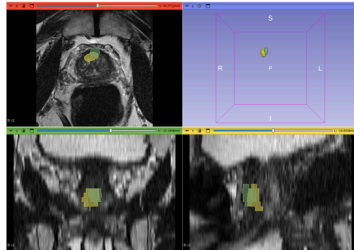
CASE 63:



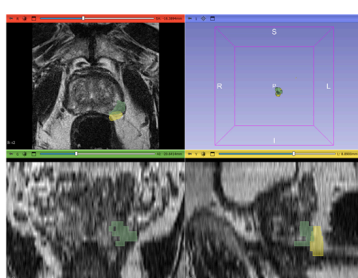
CASE 86:



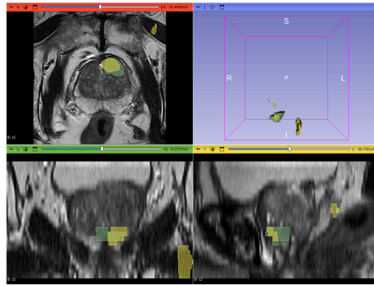
CASE 100



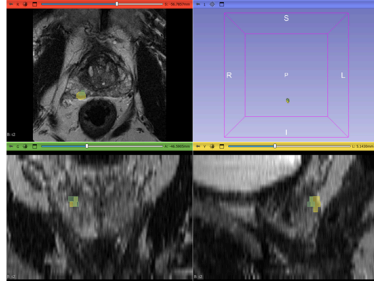
CASE 152



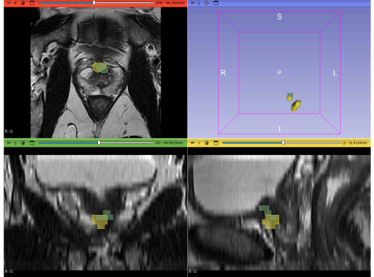
CASE 158:



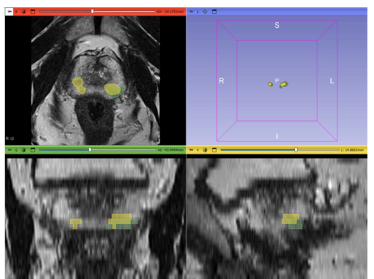
CASE 118



CASE 122



CASE 125



Sprint 5 [4/8/25 - 10/8/25]

- Development of Anatomy Segmentation Model using RRUNET3D (Modify config file) by the end of this week
- To verify the anatomy segmentation model with 3DSlicer
- To discuss with Asher by this week on how to deploy our tumor/anatomy model into AWS
- ~~[Backlog]~~ Settle Inference Pipeline and Verify the tumor prediction output with 3DSlicer again (After `monai.transforms.inverse` is applied as advised).

Sprint 5 Key Findings and Reflections

1. Prostate Anatomy Segmentation — Training Update (Anatomy_9)

Objective

Train a 3-class prostate anatomy segmentation model (background, central gland, peripheral zone) from T2-weighted MRI only, using Reader-1 annotations as ground truth.

Configuration (key points)

Data/labels: image_cols = [t2], label_cols = [t2_anatomy_reader1]; train/val/test CSVs defined.

Preprocessing: Orientation RAS; isotropic spacing $0.5 \times 0.5 \times 0.5$ mm.

Sampling: Patch-based training with RandCropPosNeg (spatial_size = 64^3 , num_samples = 8, pos:neg = 2:1).

Augmentation (subset): bias field, elastic, affine (rot/shear/translate), Gaussian/Gibbs noise, blur/sharpen, rotate/rotate90, zoom, intensity shift, contrast adjust.

Model: rrnet3d (residual + recurrent + attention), PReLU, BatchNorm, dropout 0.15; strides [2,2,2,2,2]; in_channels=1, out_channels=3.

Loss/optim: DiceCELoss (softmax, include_background=False, one-hot targets) + Novograd (lr=1e-3, weight_decay=0.01, amsgrad=True), OneCycleLR (max_lr=1e-3).

Run settings: batch_size=1, max_epochs=500, device=cuda, logs/output under anatomy_9.

Training Status & Checkpoints

Key-metric is mean Dice over foreground classes (CG, PZ).

Preliminary Results

35th Epoch: 0.3351

55th Epoch: 0.4255

205th Epoch: 0.6790

Interpretation

Dice 0.679 at epoch 205 is potentially a solid mid-training convergence for anatomy zones using T2-only input and 64^3 training patches.

Next Steps (targeted, low-risk)

- Further iterate to the 500th epoch/1000th epoch and report the DICE score
- Inference: Use full-volume sliding-window with ROI = 64^3 (or 96^3 if GPU allows), 50% overlap, and save predictions in original T2 geometry for clinical overlays.
- Patch size: Pilot 96^3 patches during training to capture more gland context (monitor VRAM).

2. Inference Pipeline Updates (Verify the tumor prediction output with 3DSlicer again (After monai inverse) is applied.)

In short:

I created 4 inference pipeline scripts and the 4th one works

inference.py: Simple single-pass inference on a center-cropped volume; saves a thresholded mask without restoring original image space.

inference2.py: Same as inference.py but resizes the mask back to the original shape; still not restored to original spatial coordinates.

inference3.py: Same as inference2.py but saves via nibabel to carry an affine; still resizes a cropped/reoriented prediction, so alignment is off.

inference4.py: Full pipeline using metadata-aware transforms, sliding-window inference, and true inversion of preprocessing to original T2 space; saves with correct affine. This yields a mask that overlays correctly.

Key differences at a glance

Cropping vs full volume

inference.py / inference2.py / inference3.py: Apply CenterSpatialCropd to $64 \times 64 \times 64$, shrinking FOV and discarding context.

inference4.py: No pre-inference crop; uses sliding-window to cover the whole volume.

Metadata handling

inference.py / inference2.py / inference3.py: Operate on tensors post-transforms and lose or ignore MONAI metadata; any later resizing is shape-only, not true space inversion. Also, SavImage or nibabel save is done without a transform inversion step.

inference4.py: Uses `EnsureTyped(..., track_meta=True)` and preserves meta for inversion.

Returning predictions to original space

inference.py: No inversion; saves cropped prediction as-is.

inference2.py: Resizes to original shape but does not undo spacing/orientation; still misaligned.

inference3.py: Resizes and saves with original affine, but still no inversion of spacing/orientation/crop; shape matches but space does not.

inference4.py: Explicitly inverts the exact preprocessing (spacing, orientation) using MONAI's `Invertd`, keyed to t2's original metadata.

Inference strategy

inference.py / inference2.py / inference3.py: Single forward pass over cropped tensor.

inference4.py: Sliding-window inference over the full preprocessed volume with overlap and gaussian blending; reconstructs a full-size prediction before inversion.

Label formation

inference.py / inference2.py / inference3.py: Softmax → select class 1 → fixed threshold 0.5.

inference4.py: Softmax → argmax via `AsDiscreted`, consistent with multi-class segmentation best practices.

Saving output

inference.py: Uses `SavImage` on a plain tensor without metadata; cannot preserve correct affine.

inference2.py: Resizes by shape and saves with `SavImage` (still lacks true space).

inference3.py: Saves with nibabel and original affine, but prediction is still in model space then resized; spatial mismatch persists.

inference4.py: Inverts to original T2 space first, then saves with affine from metadata (or original affine as fallback); additionally guards with a nearest-neighbor resize if shape still differs.

Why did inference4.py produces a correct overlay

It preserves transform metadata and uses `Invertd` to undo spacing and orientation on the prediction, referencing the original t2 image's metadata.

It avoids destructive pre-inference cropping by using a sliding-window inferer, predicting the whole volume and then inverting to native space.

It writes the output with the correct affine. If any residual shape mismatch remains after inversion, it applies a nearest-neighbor resize for label integrity, ensuring exact voxel grid and affine match.

Net effect: the final mask is in the same coordinate system and voxel grid as the raw T2, so clinical viewers can overlay it correctly.

inference4.py: Full pipeline using metadata-aware transforms, sliding-window inference, and true inversion of preprocessing to original T2 space; saves with correct affine. This yields a mask that overlays correctly. metadata-aware pipeline (`ConcatItemsd` + `EnsureTyped`), sliding-window over full volume, `Activationsd`+`AsDiscreted`, and `Invertd` to original T2 space; saved with correct affine. This is why overlay works.

What did I verify with?

2 Patient Cases with Tumor Ground Truth (Case 51 and 63)

The tumor prediction output overlapped with T2-Reader1 ground truth segmentation.

Will upload the slicer3d screenshots here soon.

Next steps:

MORE Sanity checks, at least 10 cases to be reviewed.

Sprint 4

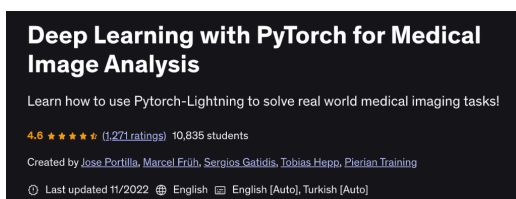
1. Set up time to populate stories to clear
2. Select a date to meet every week amongst Anson and Asher - every **Thursday**
3. Every week find time to update Sufyan on whole progress in the BOARD - **Thursday**
4. Get Laptop to start working. Technically own internet laptop also can.

Sprint 3 (Training)

Check the image sizes, image sizes used to train the model, and how much computing

Get training script up

Sprint 2 (Medical Image Analysis + Going through Sufyan's Shared Project links Prostate158 and MONAI)



Went through the above course, though my focus mainly wasn't on pytorch-lightning but more on how to apply DL on Medical Image.

Summary of the udemy course:

What you'll learn

- ✓ Learn how to use NumPy
- ✓ Foundations of Medical Imaging
- ✓ Creating Artificial Neural Networks with PyTorch
- ✓ Visualize the decision of a CNN
- ✓ Automatic Cancer Segmentation
- ✓ Learn classic machine learning theory principals
- ✓ Data Formats in Medical Imaging
- ✓ Use PyTorch-Lightning for state of the art training
- ✓ 2D & 3D data handling

The course included pytorch basics and CNN. Key Learnings were understanding what kind of Medical Imagings exists (X-RAY/CT/MRI/PET) and why the need to preprocess them differently. Also learnt the data formats and preprocessing methods in medical imaging such as DICOM and NIfTI, and how to use libraries like pydicom and nibabel to handle the medical images.

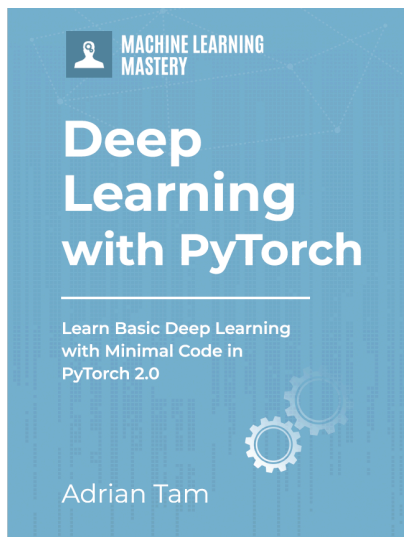
Also managed to immerse in guided projects such as Pneumonia-Classification, Cardiac-detection, Atrium Segmentation, Lung Tumor Segmentation and 3D Liver and Liver Tumour Segmentation.

Went Through Prostate-158 Project (UNET), and replacing UNET with MONAI RRUNET3D Model .

Key **bottleneck**: Training Notebook kept crashing and crashing Sprint 1 (Learn Deep Learning from scratch)

Learning from 2 Sources:

- DataCamp
-



Learning Pointers:

1. Introduction to Pytorch (Advantages and Limitations, Applications)
2. Manipulating tensors in Pytorch (Creating/Checking/Manipulating Tensors/Tensors Functions)
3. Autograd in PyTorch
4. Deep Learning Crash Course (What are MLPs, Neurons: Weights and Activations, Training Networks)
5. Multilayer perceptron building blocks in Pytorch (NN Models, Model Inputs, Layers, Activations, and Layer Properties, Model Training and Inference)
6. Basics of creating NN in Pytorch (Point 1 - 5)
7. How to create training loop for models with statistics and progress bar from tqdm library
8. Evaluation of models (data splitting/k-fold cv)
9. How I learnt to build Binary/Multi-class classification model in PyTorch
10. Techniques for better DL models (Save and load Pytorch models, use Activation Functions, Loss Functions)
11. Learnt about Dropout regularization to combat overfitting and improve model accuracy
12. Learning Rate and Momentum in Pytorch
13. Use case of DataLoader and DataSet
14. Using Pytorch DL models with scikit-learn
15. Hyperparameter Tuning
16. Managing Training Process with Checkpoints and Early Stopping
17. Pytorch Model visualization
18. Understand model behavior during training by visualizing metrics
19. From MLP to CNN to RNN
20. Building CNN
21. Mnist Project recap with LeNet5 model in Pytorch
22. Using 3D Slicer to visualize Medical Image (**Basics only**)
23. Basics of Image Segmentation and the models available